

Chapter 1.3 Random Variables

Rick Durrett Probability: Theory and Examples

Hu Jingwu

Built from the local Chapter 1 LLMviki flash cards.

Roadmap for 1.3

Learning goal

Prove measurability quickly using generators and closure under limits/composition.

1. Random variables as measurable maps.
2. Generator checking and inverse images.
3. Arithmetic, composition, and limits of random variables.
4. Semicontinuity and discontinuity sets.
5. $\sigma(X)$ factorization: when $Y = f(X)$.

Random Variables as Measurable Maps

Definition

A random variable is a measurable map

$$X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

Equivalently,

$$\{X \in B\} \in \mathcal{F} \quad \text{for every } B \in \mathcal{B}(\mathbb{R}).$$

Pitfall

A random variable is not the same thing as its distribution. Different maps may have the same law.

LLMwiki: Random variables as measurable maps; ID: durrett_ch1_measurable_map

Checking Measurability on Generators

Theorem

If $\mathcal{S} = \sigma(\mathcal{A})$, then to prove $X : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{S})$ is measurable, it is enough to check

$$X^{-1}(A) \in \mathcal{F} \quad \text{for every } A \in \mathcal{A}.$$

LLMwiki: [Checking measurability on generators](#); ID: durrett_ch1_measurability_generator

Composition of Measurable Maps

Theorem

If X is measurable and f is measurable, then $f(X)$ is measurable.

LLMwiki: Composition of measurable maps; ID: durrett_ch1_composition_measurable

Arithmetic of Random Variables

Theorem

Finite sums, products, maxima, minima, and continuous functions of random variables are random variables.

- Vector trick: first prove (X, Y) is measurable.
- Then compose with continuous maps such as $(x, y) \mapsto x + y$ or $(x, y) \mapsto xy$.
- Direct trick for sums: use rational cuts.

LLMwiki: [Arithmetic of random variables](#); ID: durrett_ch1_measurable_arithmetic

Limits Preserve Measurability

Theorem

If (X_n) are measurable real functions, then

$$\sup_n X_n, \quad \inf_n X_n, \quad \limsup_n X_n, \quad \liminf_n X_n$$

are measurable. Any pointwise limit of measurable functions is measurable.

LLMwiki: Limits preserve measurability; ID: durrett_ch1_limits_measurable

$\sigma(X)$ Factorization

Fact

A random variable Y is $\sigma(X)$ -measurable if and only if there is a Borel measurable $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$Y = f(X).$$

- Easy direction: inverse images compose.
- Hard direction: approximate Y by simple functions or dyadic bins.

LLMviki: Functions measurable with respect to $\sigma(X)$; ID: durrett_ch1_sigma_x_factorization

Exercise 1.3.1

Show that if $\mathcal{A}$ generates $\mathcal{S}$, then

$$X^{-1}(\mathcal{A}) \equiv \{\{X \in A\} : A \in \mathcal{A}\}$$

generates

$$\sigma(X) = \{\{X \in B\} : B \in \mathcal{S}\}.$$

LLMwiki: [Checking measurability on generators](#); ID: durrett_ch1_measurability_generator

Exercise 1.3.1: Proof Skeleton

LLMwiki: Checking measurability on generators; ID: durrett_ch1_measurability_generator

Exercise 1.3.2

Prove Theorem 1.3.6 when $n = 2$ by checking directly that

$$\{X_1 + X_2 < x\} \in \mathcal{F}.$$

LLM**wiki:** Rational cut proof for sum measurability; ID: exercise_method_rational_cut_sum_measurability

Exercise 1.3.2: Proof Skeleton

LLMwiki: Rational cut proof for sum measurability; ID: exercise_method_rational_cut_sum_measurability

Exercise 1.3.3

Show that if f is continuous and $X_n \rightarrow X$ almost surely, then

$$f(X_n) \rightarrow f(X)$$

almost surely.

LLMwiki: Composition of measurable maps; ID: durrett_ch1_composition_measurable

Exercise 1.3.3: Proof Skeleton

LLMwiki: Composition of measurable maps; ID: durrett_ch1_composition_measurable

Exercise 1.3.4

- (i) Show that a continuous function from $\mathbb{R}^d$ to $\mathbb{R}$ is measurable from $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.
- (ii) Show that $\mathcal{B}(\mathbb{R}^d)$ is the smallest sigma-field that makes all continuous functions measurable.

LLMwiki: Composition of measurable maps; ID: durrett_ch1_composition_measurable

Exercise 1.3.4: Proof Skeleton

LLMwiki: Checking measurability on generators; ID: durrett_ch1_measurability_generator

Exercise 1.3.5

A function f is lower semicontinuous if

$$\liminf_{y \rightarrow x} f(y) \geq f(x),$$

and upper semicontinuous if $-f$ is lower semicontinuous. Show that f is lower semicontinuous if and only if $\{x : f(x) \leq a\}$ is closed for each $a \in \mathbb{R}$, and conclude that semicontinuous functions are measurable.

LLMwiki: Lower semicontinuity via closed sublevel sets; ID:
`exercise_method_closed_sublevel_semicontinuity`

Exercise 1.3.5: Proof Skeleton

LLMwiki: Lower semicontinuity via closed sublevel sets; ID:
`exercise_method_closed_sublevel_semicontinuity`

Exercise 1.3.6

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be arbitrary and define

$$f^\delta(x) = \sup\{f(y) : |y - x| < \delta\}, \quad f_\delta(x) = \inf\{f(y) : |y - x| < \delta\}.$$

Show that f^δ is lower semicontinuous and f_δ is upper semicontinuous. Let

$$f^0 = \lim_{\delta \downarrow 0} f^\delta, \quad f_0 = \lim_{\delta \downarrow 0} f_\delta.$$

Conclude that the set of discontinuities of f is $\{f^0 \neq f_0\}$ and is measurable.

LLMwiki: Discontinuity set from upper and lower local envelopes; ID:

`exercise_method_oscillation_discontinuity_set`

Exercise 1.3.6: Proof Skeleton

LLMwiki: Discontinuity set from upper and lower local envelopes; ID:
`exercise_method_oscillation_discontinuity_set`

Exercise 1.3.7

A function $\phi : \Omega \rightarrow \mathbb{R}$ is simple if

$$\phi(\omega) = \sum_{m=1}^n c_m \mathbf{1}_{A_m}(\omega), \quad A_m \in \mathcal{F}.$$

Show that the class of $\mathcal{F}$ -measurable functions is the smallest class containing the simple functions and closed under pointwise limits.

LLMwiki: Limits preserve measurability; ID: durrett_ch1_limits_measurable

Exercise 1.3.7: Proof Skeleton

LLMwiki: Limits preserve measurability; ID: durrett_ch1_limits_measurable

Exercise 1.3.8

Use the previous exercise to conclude that Y is measurable with respect to $\sigma(X)$ if and only if

$$Y = f(X)$$

for some measurable function $f : \mathbb{R} \rightarrow \mathbb{R}$.

LLMwiki: Functions measurable with respect to $\sigma(X)$; ID: durrett_ch1_sigma_x_factorization

Exercise 1.3.8: Proof Skeleton

LLMwiki: Functions measurable with respect to $\sigma(X)$; ID: durrett_ch1_sigma_x_factorization

Exercise 1.3.9

Give a constructive proof of the previous result. Note that

$$\{\omega : m2^{-n} \leq Y(\omega) < (m+1)2^{-n}\} = \{X \in B_{m,n}\}$$

for some Borel set $B_{m,n}$, and define $f_n(x) = m2^{-n}$ for $x \in B_{m,n}$. Show that $f_n(x) \rightarrow f(x)$ and $Y = f(X)$.

LLMwiki: Constructive representation of sigma(X)-measurable functions; ID:

`exercise_method_constructive_sigma_x_representation`

Exercise 1.3.9: Proof Skeleton

LLMwiki: Constructive representation of $\sigma(X)$ -measurable functions; ID:
`exercise_method_constructive_sigma_x_representation`

Checklist: How to Attack 1.3 Problems

1. To prove measurability, reduce to a generator.
2. For sums, use rational cuts when a direct proof is requested.
3. For limits, rewrite events using countable unions/intersections.
4. For semicontinuity, use closed sublevel sets or open superlevel sets.
5. For $\sigma(X)$ -measurability, think “information carried by X ” and prove $Y = f(X)$.

LLMwiki: [Chapter 1.3 flash cards and exercise method cards](#)

Mini Practice

1. Prove that $\max(X, Y)$ is measurable using rational cuts.
2. If X_n are random variables, show $\{\lim_n X_n \text{ exists}\} \in \mathcal{F}$.
3. Show every continuous $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is Borel measurable.
4. If Y is $\sigma(X)$ -measurable and g is Borel, explain why $g(Y)$ is $\sigma(X)$ -measurable.